

MUHAMMAD AKMAL

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SUMMARY

Detail-oriented Computer Science graduate with hands-on expertise in Python, Machine Learning, and Deep Learning, developed through academic coursework and applied personal projects. Experienced in building AI-driven applications, including voice emotion analysis and movie recommendation systems, using frameworks such as TensorFlow, Keras, FastAPI, and Scikit-Learn. Skilled in data preprocessing, model development, and end-to-end application deployment, with a strong analytical mindset and a passion for artificial intelligence, data science, and continuous learning. Seeking to apply technical expertise and problem-solving abilities as a Junior ML Engineer.

EDUCATION

University of Education Lahore

Bachelor of Science in Computer Science

2022 – 2026

Lahore, Pakistan

Aspire Group of Colleges

Intermediate – F.Sc. Pre-Engineering

2020 – 2022

Lahore, Pakistan

PROJECTS

Emotion-Aware Voice Processing for Customer Service

Python | FastAPI | React | MachineLearning | Librosa

- Designed and implemented an AI-driven speech analysis platform using Whisper and CNN+BiLSTM models for transcription, emotion detection, and audio processing.
- Developed and deployed scalable FastAPI services to handle audio processing workflows and deliver real-time transcription and emotion analysis results.
- Trained deep learning models on RAVDESS and TESS datasets while applying advanced audio processing techniques to improve classification performance.
- Created a responsive React-based dashboard to present transcription outputs, emotion predictions, and processed audio insights for end users.

Movie Recommendation System

Python | MachineLearning | Streamlit

- Designed and implemented a machine learning-based recommendation engine to generate personalized movie suggestions from user preferences and content attributes.
- Performed data preprocessing and analysis on movie datasets to structure inputs for model training.
- Built and deployed an interactive Streamlit web application to deliver real-time recommendations through a user-friendly interface.

TECHNICAL SKILLS

Programming Languages: Python

Machine Learning & AI: Machine Learning, Deep Learning, TensorFlow, Keras, Scikit-Learn

Data Science: Data Analysis, Data Preprocessing, Pandas, NumPy

Frameworks & Tools: FastAPI, React, Streamlit, Librosa

Version Control: Git, GitHub

CERTIFICATIONS

- FOC Certificate – Lincoln Corner (A Project of the U.S. Embassy Pakistan)
- Crash Course on Python
- Use AI as a Creative or Expert Partner

LANGUAGES

English | Urdu